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sS	>	hH	
$2h(H-10)$	>	hH	$s=2h$ and $S=H-10$
$2hH-20h$	>	hH	distributive property
$2hH$	>	$hH+20h$	add $20h$ to both sides
hH	>	$20h$	subtract hH from both sides
H	>	20	divide both sides by $h>0$

Now reverse the steps.

H	>	$\frac{62}{3}$	derived earlier
H	>	20	$\frac{62}{3}>20$
hH	>	$20h$	multiply both sides by $h>0$
$2hH$	>	$hH+20h$	add hH to both sides
$2hH-20h$	>	hH	subtract $20h$ from both sides
$2h(H-10)$	>	hH	factor
sS	>	hH	$s=2h$ and $S=H-10$

Thus, the revenue from the sale of softcover books was greater than the revenue from the sale of hardcover books.

The correct answer is C; both statements together are sufficient.

•

At a certain refreshment stand, all hot dogs are the same price, and all sodas are the same price. What is the total price of 3 hot dogs and 2 sodas at the refreshment stand?

1. The total price of 5 sodas at the stand is less than the total price of 2 hot dogs.
2. The total price of 9 hot dogs and 6 sodas at the stand is $21.

Algebra Applied Problems

Let $\$H$ be the price of each hot dog and $\$S$ be the price of each soda. We are to determine the value of $3H+2S$.

1. Given that $5S<2H$, the value of $3H+2S$ could be 7.5 (if $H=2$ and $S=0.75$), and the value of $3H+2S$ could be 11 (if $H=3$ and $S=1$); NOT sufficient.
2. Given that $9H+6S=21$ and recognizing that the left-hand side of this equation is 3 times $3H+2S$ (what we want the value of), divide both sides of this equation by 3 to get $3H+2S=7$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

If a certain school will buy computers at a fixed price of x dollars each and computer desks at a fixed price of y

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If a certain school will buy computers at a fixed price of x dollars each and computer desks at a fixed price of y dollars each, how many dollars will the school pay for each computer?

1. The school will pay a total of $2,200 if it buys 2 computers and 1 computer desk at these fixed prices.
2. The school will pay a total of $5,400 if it buys 5 computers and 2 computer desks at these fixed prices.

Algebra Simultaneous Equations

1. Given that $2x + y = 2{,}200$, by choosing different values for y and solving for x, it follows that more than one value of x is possible, and hence the value of x cannot be determined. For example, $x = 1{,}000$ is possible (choose $y = 200$), and $x = 900$ is possible (choose $y = 400$); NOT sufficient.
2. Given that $5x + 2y = 5{,}400$, by choosing different values for y and solving for x, it follows that more than one value of x is possible, and hence the value of x cannot be determined. For example, $x = 1{,}000$ is possible (choose $y = 200$), and $x = 920$ is possible (choose $y = 400$); NOT sufficient.

Taking (1) and (2) together, we have $2x + y = 2{,}200$ and $5x + 2y = 5{,}400$. These two equations can be solved simultaneously to obtain a unique value of x. This can be done by solving for y in terms of x in the first equation to get $y = 2{,}200 - 2x$, then replacing y with $2{,}200 - 2x$ in the second equation to get an equation, namely the linear equation $5x + 2(2{,}200 - 2x) = 5{,}400$, that can be solved for a unique value of x. Although we do not have to find the actual value of x, for those interested, solving this equation gives $x = 1{,}000$.

Alternatively, these two equations, when graphed in the standard xy-coordinate plane, represent two lines having different slopes (hence the lines are not parallel), and therefore the two lines have exactly one point in common. The x-coordinate of that common point is the unique value of x that satisfies both equations.

The correct answer is C; both statements together are sufficient.

•

An initial investment of $10,000 was deposited in a bank account one year ago, and additional deposits were made during the year. If no withdrawals were made, what was the total amount of interest earned on this account during the year?

1. The additional deposits during the year totaled $5,000.
2. The account earned interest at the annual rate of 6 percent compounded quarterly.

Arithmetic Applied Problems

Determine the interest earned in one year by an initial investment of $10,000 with additional deposits, but no withdrawals, during the year.

1. Given that the additional deposits total $5,000, it is not possible to determine the interest earned by both the initial investment and the additional deposits together without information about the interest rate and when during the year the additional deposits were made; NOT sufficient.
2. Given that the annual interest rate is 6% compounded quarterly, it is not possible to determine the interest earned by both the initial investment and the additional deposits together without information about the amount of the additional deposits and when during the year the additional deposits were made; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the interest earned by both the initial investment and the additional deposits together without information about when during the year the additional deposits were made. For example, if one deposit of $5,000 were made after 6 months, then the interest for the year would be more than it would have been had the $5,000 been deposited after 9 months.

The correct answer is E; both statements together are still not sufficient.

•

A poplar tree was 3 feet high when it was planted on January 1, 1970. During what year did it pass the height of 20 feet?

1. On January 1, 1973, it was 24 feet high.
2. It doubled its height during each year.

Arithmetic Series and Sequences

1. Given that the tree was 24 feet high at the beginning of 1973, it is not possible to determine during which year the tree passed the height of 20 feet.

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the tree passed the height of 20 feet.

year	beginning height (ft)	ending height (ft)	feet grown (ft)
1970	3	15	12
1971	15	21	6
1972	21	24	3
1973	24	30	6

year	beginning height (ft)	ending height (ft)	feet grown (ft)
1970	3	6	3
1971	6	12	6
1972	12	24	12
1973	24	30	6

The first table shows that the tree could have passed the height of 20 feet during 1971 and the second table shows that the tree could have passed the height of 20 feet during 1972; NOT sufficient.

2. Given that the tree doubled its height during each year, the tree would have been 6 feet high at the beginning of 1971, 12 feet high at the beginning of 1972, and 24 feet high at the beginning of 1973. Therefore, the tree would have passed the height of 20 feet during 1972; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

Which weighs more, a cubic unit of water or a cubic unit of liquid X?

1. A cubic unit of water weighs more than $\frac{1}{3}$ cubic unit of liquid X.
2. A cubic unit of liquid X weighs less than 3 cubic units of water.

Algebra Inequalities

Determine which is greater: the weight of a cubic unit of water, represented by W, or a cubic unit of Liquid X, represented by X.

1. Given that $W > \frac{1}{3}X$, it is not possible to determine which of W and X is greater. For example, if $W = 4$ and $X = 9$, then $W > \frac{1}{3}X$ and X is greater than W, but if $W = 10$ and $X = 9$, then $W > \frac{1}{3}X$ and W is greater than X; NOT sufficient.
2. Given that $X < 3W$, it is not possible to determine which of W and X is greater. For example, if $X = 9$ and $W = 4$, then $X < 3W$ and X is greater than W, but if $X = 9$ and $W = 10$, then $X < 3W$ and W is greater than X; NOT sufficient.

Taking (1) and (2) together gives no more information than (1) or (2) alone since the same examples used to show that (1) is not sufficient also show that (2) is not sufficient.

The correct answer is E; both statements together are still not sufficient.

•

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What were the individual prices of the vases that an antique dealer bought at store X?

1. The antique dealer bought exactly 3 vases at store X.
2. The antique dealer's total bill at store X was $225.

Arithmetic Applied Problems

Determine the individual prices of the vases.

1. Given that there are 3 vases, it is not possible to determine the individual prices because no information about prices is known or can be determined; NOT sufficient.
2. Given that the total bill was $225, it is not possible to determine the individual prices because neither the number of vases nor whether the vases are identically or otherwise priced is known or can be determined; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the individual prices of the vases. For example, the prices of the 3 vases could be $200, $20, and $5 for a total bill of $225. However, the prices of the 3 vases could also be $100, $100, and $25 for a total bill of $225.

The correct answer is E; both statements together are still not sufficient.

•

Was the average (arithmetic mean) sale price of a new home in region R last month at least $100,000?

1. Last month the median sale price of a new home in region R was at least $100,000.
2. Last month the sale prices of new homes in region R ranged from $75,000 to $150,000.

Arithmetic Statistics

1. Given that the median price was at least $100,000, the following two examples show that it cannot be determined whether the average price was at least $100,000.

   Example 1: Average price is greater than $100,000

$75,000	**$100,000**	$150,000
$100,000		$150,000
$100,000		$150,000

   The median of these 7 prices is $100,000 and the average of these prices is greater than $100,000, since the sum of these 7 prices is 7($100,000) + (−$25,000 + $50,000 + $50,000 + $50,000), which is greater than 7($100,000).

   Example 2: Average price is less than $100,000

$75,000	**$100,000**	$150,000
$100,000		$150,000
$100,000		$150,000

   The median of these 7 prices is $100,000 and the average of these prices is less than $100,000, since the sum of these 7 prices is 7($100,000) + (−$25,000 − $25,000 − $25,000 + $50,000), which is less than 7($100,000); NOT sufficient.

2. Given that the prices ranged from $75,000 to $150,000, the same examples above show that it cannot be determined whether the average price was at least $100,000; NOT sufficient.

Taking (1) and (2) together, it cannot be determined whether the average price was at least $100,000 because the two examples above each satisfy both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

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•

If the capacity of tank X is less than the capacity of tank Y and both tanks begin to fill at the same time, which tank will be filled first?

1. Tank X is filled at a constant rate of 1.5 liters per minute.
2. Tank Y is filled at a constant rate of 120 liters per hour.

Arithmetic Applied Problems

Determine which tank, X or Y, will be filled first if X has less capacity than Y and they start filling at the same time.

1. Given that X fills at a rate of 1.5 liters per minute, which is equivalent to 90 liters per hour, it is not possible to determine which tank will be filled first because no information is given about the rate at which Y fills or about how much larger in capacity Y is than X; NOT sufficient.
2. Given that Y fills at a rate of 120 liters per hour, it is not possible to determine which tank will be filled first because no information is given about the rate at which X fills or about how much larger in capacity Y is than X; NOT sufficient.

Taking (1) and (2) together, if the capacity of X is 90 liters and the capacity of Y is 200 liters, then X will be filled in 1 hour but Y will be only 60% filled in 1 hour. Therefore, X will be filled first. However, if the capacity of X is 90 liters and the capacity of Y is 100 liters, then Y will be filled in $\frac{5}{6}$ hours while X will take a full hour to be filled. Thus, Y will be filled first.

The correct answer is E; both statements together are still not sufficient.

•

At a certain company, 30 percent of the employees live in City R. If 25 percent of the company's employees live in apartments in City R, what is the number of the employees who live in apartments in City R?

1. Of the employees who live in City R, 6 do <u>not</u> live in apartments.
2. Of the employees, 84 do <u>not</u> live in City R.

Arithmetic Percents

Determine how many employees of a certain company live in apartments in City R, where 30% of the employees live in City R and 25% of the employees live in apartments in City R. Let T represent the total number of employees at the company.

1. Given that 6 of the employees who live in City R do not live in apartments, it follows that $6 = (0.30 - 0.25)T$. Thus $0.05T = 6$, $T = 120$, and the number of employees who live in apartments in City R can be determined; SUFFICIENT.
2. Given that 84 employees do not live in City R, it follows that $(1 - 0.3)T = 84$, from which it follows that $T = 120$ and the number of employees who live in apartments in City R can be determined; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

What was Mary's average (arithmetic mean) score on 4 tests?

1. Her average (arithmetic mean) score on 3 of the tests was 97.
2. Her score on one of the tests was 96.

Arithmetic Statistics

Since the average of the 4 scores is equal to the sum of the 4 scores divided by 4, it follows that the average of the 4 scores can be determined if and only if the sum of the 4 scores can be determined.

1. Given that the sum of 3 of the scores was 3(97), it is not possible to determine the sum of the 4 scores, since different values for the remaining score are possible and those different values correspond to different values for the sum of the 4 scores; NOT sufficient.
2. Given that one of the scores was 96, it is not possible to determine the sum of the 4 scores, since different values for the sum of the remaining 3 scores are possible and those different values correspond to different values for

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for the sum of the remaining 3 scores are possible and those different values correspond to different values for the sum of the 4 scores; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the sum of the 4 scores. For example, the scores could be 96, 97, 98, 10 (the first 3 listed scores have an average of 97 and one of the scores is 96), which have an average that is less than 96, and the scores could be 96, 97, 98, 99 (the first 3 listed scores have an average of 97 and one of the scores is 96), which have an average that is greater than 96.

The correct answer is E; both statements together are still not sufficient.

•

Is there a causal relationship between smoking and lung cancer?

1. Research consistently shows a strong correlation between smoking and the development of lung cancer.
2. Some medical researchers support a proposed mechanism by which smoking could cause lung cancer.

Inference

1. Research-based evidence has consistently shown a high positive correlation between smoking and lung cancer. A strong positive correlation between two factors P and Q indicates that there is a similar pattern of variation in data for P and data for Q (the degree of similarity can vary). For example, long-term data might show that as smoking increases in a population, the data regarding the incidence of lung cancer increases in tandem. Over several years, if the data regarding smoking in a population decreases, the data for lung cancer might also decrease. In both cases, a positive correlation occurs. But such a statistical pattern, by itself, can, at best, *suggest* some *association* or *dependency*, direct or indirect, between the two factors smoking and lung cancer. But correlation evidence, by itself, provides no proof of a causal relationship; NOT sufficient.
2. The information provided is insufficiently specific to sustain a claim that smoking is causally related to lung cancer. What (2) indicates is a *hypothesis* proposed by some researchers, but no information is provided to indicate confirmation of that hypothesis; NOT sufficient.

The correct answer is E; both statements together are still not sufficient.

•

A research project has a successful outcome if its research is reported in a blind peer-reviewed academic publication. Can interdisciplinary collaborations on research projects at least sometimes produce a successful outcome?

1. Some interdisciplinary research groups experience conflict and rivalry.
2. A peer-reviewed and well-regarded interdisciplinary review of published papers on urban greening trends found that there was little if any evidence that planting more trees in an urban area significantly improved air quality there.

Inference

1. The information regarding the sometimes dysfunctional social dynamics of research groups is not sufficient to show that research groups with such experiences always fail to produce successful research outcomes. So the information given here is not sufficient alone to indicate a yes or a no answer to the question posed; NOT sufficient.
2. The information gives us an example of one interdisciplinary collaboration on research that reported in a blind peer-reviewed publication its review of research studies on urban greening. The review found that there was insufficient evidence to show that planting trees in urban areas had a significant impact on air quality in such areas. Even though the result of its review was negative, its review of scientific publications on the issue was approved in a blind peer review and published. Therefore (2), in combination with the stimulus information, is sufficient to provide at least one example of a successful outcome of an interdisciplinary research project. This, in turn, is sufficient to show that interdisciplinary research groups *can* produce a successful outcome.; SUFFICIENT

The correct answer is B; statement 2 alone is sufficient.

•

It has been proposed that teenagers aged 10 to 15 years be restricted to less than 2 hours per day engaging with social media. Would a significant number of teens aged 10 to 15 years get an overall developmental or health benefit from such a restriction?

1. A peer-reviewed study indicates a 23% increase in the incidence of chronic anxiety or depression among

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1. A peer-reviewed study indicates a 23% increase in the incidence of chronic anxiety or depression among teenagers aged 10 to 15 years who average 2 or more hours per day engaging with social media.
2. Certain kinds of engagement with social media averaging 2 or more hours per day by teenagers aged 10 to 15 years lead, in about 28% of cases, to meaningful friendships and social and emotional learning, both of which are valuable for development at those ages.

Inference

1. This statement cites a study that found an association between social-media use by teenagers 10 to 15 years and an increase in the incidence of chronic anxiety or depression among 23% of the minors observed in the study. A key word here is *association*: It should be noted that this does not prove a cause-effect relationship, even if the study could motivate further investigation to see whether a causal mechanism could be identified. Even if a causal mechanism were identified, it could turn out that the use of social media was at least in part an *effect* of chronic anxiety or depression: Some teenagers might resort to social media use as a kind of escape mechanism. We should conclude that the information provided (even if the validity of the study is assumed) does not provide sufficient information to show definitively that the proposed restriction would result in developmental or health benefits for minors aged 10 to 15 years. The conclusion is that (1) is not sufficient alone; NOT sufficient.
2. This statement indicates that use of social media by teens aged 10 to 15 years results in significant developmental benefits for 28% of teens in that age group. In other words, it indicates that social-media use by these teens *causally contributes* to a developmentally valuable effect. This implies that restriction of their social media use risks depriving a significant number of teens of the benefits resulting from their social-media use. So we can conclude that (2) is sufficient alone to provide a negative answer to the question posed; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

A milk vendor mixes water with milk and sells the mixture at the same price per liter as if it were undiluted milk. The selling price per liter of the mixture is the vendor's cost per liter of the milk plus a markup of x%. The water costs the vendor nothing. If the vendor gets a 50% profit on the sale of the mixture, what is the value of x?

1. If the vendor mixes half the intended quantity of water and sells every liter of the mixture at the cost price per liter of the undiluted milk, the vendor will get a 10% profit.
2. The concentration of milk in the mixture after adding water is 5/6.

Evaluate

The task is to ascertain whether enough information has been provided to determine the value of x.

1. In this hypothetical scenario, we are asked to imagine that the amount of zero-cost water added to the milk is only half as much as was actually used, based on the information given in the stimulus. The profit in this hypothetical scenario is 10%, resulting from selling the mixture at the vendor's **cost price** per liter of milk (i.e., without any profit markup). Therefore, the 10% profit per liter in this hypothetical scenario consists, in effect, of selling zero-cost water at the cost price of milk. We can infer from this that the ratio of water to milk in the mixture was 1:10. When 20% **zero-cost water** is added, it can be inferred from (1) that the ratio of water to milk in the mixture is 2:10. This would result in a 20% profit if every liter of the mixture is sold at the cost price per liter of milk. But suppose that for retail sale, there is a markup of x% over cost on every liter of the mixture. We are told in the preliminary information that a total profit of 50% accrues. The 30% difference between the 20% profit and the 50% profit results from the markup of x%. It is 30% of the cost price of milk. Note that the question concerns the value of x, not the **% profit** resulting from the markup of x%. The information in (1) and the preliminary information is sufficient to determine that $x = 25$. In other words, the markup on the vendor's cost for the milk is 25%. Some simple algebra will show that the x% markup contributes 30% to the 50% profit; SUFFICIENT.
2. This information indicates that the water in each liter of the mixture is 1/5 of the amount of milk. So, pricing the zero-cost water as if it were milk accounts for 20% out of the 50% profit. The markup of x% on the whole mixture accounts for the remaining 30%. Therefore, the information in (2) is sufficient by itself to determine the value of x; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

Does it benefit a company overall to invest in employee training?

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1. Companies that invest in comprehensive employee training programs often observe improved job performance among their workforce.
2. Well-trained employees are likely to excel in their roles and take the company forward.

Inference

1. Even if companies that invest in comprehensive employee training programs **often** observe improved job performance among their workforce, many such companies might not observe any improved job performance. And even for those that do, the improvements might be too minimal to make it beneficial overall for the company to pay the costs for the training programs. Furthermore, even if investing specifically in a comprehensive employee training program did benefit a company overall, investing in most other types of employee training might not be. Thus, whether it benefits an unspecified company overall to invest in an unspecified type of employee training cannot be determined from (1) alone; NOT sufficient.
2. A company's investment in employee training may fail to produce any well-trained employees. And even if well-trained employees are **likely** to excel in their roles and take the company forward, many well-trained employees may not do so. Furthermore, the hazy benefits of some employees "excelling" and taking the company "forward" might not be adequate to justify the costs of employee training. Thus, whether it benefits a company overall to invest in employee training cannot be determined from (2) alone; NOT sufficient.

Even if (1) and (2) are both true, we still can't determine whether the benefits of improved job performance and of more employees excelling and moving a particular company forward would make the training benefit the company overall, considering the costs of the training. Nor can we even determine whether a specific employee training program would yield any such benefits at all.

The correct answer is E; both statements together are still not sufficient.

•

Can regular exercise reduce the risk of chronic diseases?

1. Regular exercise is often associated with better physical health.
2. Engaging in regular physical activity contributes to overall well-being.

Inference

1. For all we know from (1), regular exercise may often be associated with better physical health simply because people in better physical health are often more inclined to exercise. And even if regular exercise promotes better physical health in some respects, it might not reduce the risk of chronic diseases specifically. Thus, whether regular exercise can reduce the risk of chronic diseases cannot be determined from (1) alone; NOT sufficient.
2. Even if engaging in regular physical activity contributes to overall well-being, it may not reduce the risk of chronic diseases specifically. Thus, whether regular exercise can reduce the risk of chronic diseases cannot be determined from (2) alone; NOT sufficient.

Taking (1) and (2) together, even if regular exercise is both associated with better physical health and contributes to overall well-being, it may not reduce the risk of chronic diseases specifically.

The correct answer is E; both statements together are still not sufficient.

•

Does effective advertising require a higher budget expense than ineffective advertising?

1. Companies that allocate higher budgets to advertising tend to experience increased sales revenue.
2. Effective advertising can drive higher consumer demand.

Inference

1. Even if companies that allocate higher budgets to advertising tend to also have increased sales revenue, that might mean only that their increased sales revenue has given them more money to spend on their ineffective advertising. Furthermore, even those companies allocating high budgets to **effective** advertising might be doing so inefficiently; they might be able to cut their advertising expenses drastically without reducing their advertising effectiveness. Thus, whether effective advertising **requires** a higher budget expense than ineffective advertising cannot be determined from (1) alone; NOT sufficient.
2. Even if effective advertising can drive higher consumer demand, it might not need to cost much. Thus, whether effective advertising requires a higher budget expense than ineffective advertising cannot be determined from (2)

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effective advertising requires a higher budget expense than ineffective advertising cannot be determined from (2) alone; NOT sufficient.

Taking (1) and (2) together, even if companies that allocate higher budgets to advertising also tend to have increased sales revenue, and even if effective advertising can drive higher consumer demand, many companies with higher budgets and increased sales revenue may nonetheless be advertising inefficiently; they might be able to cut their advertising expenses drastically without reducing advertising effectiveness. If so, effective advertising may not require a higher budget expense than ineffective advertising.

The correct answer is E; both statements together are still not sufficient.

Questions 39 to 75 — Difficulty: Medium

39. If a certain store purchased a computer game and then sold it for \$50, was the store's profit from the purchase and sale of the game less than \$8?

    1. The store's profit from the purchase and sale of the game was less than 25 percent of the amount for which the store purchased the game.
    2. The store's profit from the purchase and sale of the game was greater than 10 percent of the amount for which the store purchased the game.

    Arithmetic Applied Problems

    Let $\$p$ be the store's purchase price of the computer game. Then the store's profit was $\$(50-p)$. We are to determine whether $50-p<8$, or equivalently, whether $p>42$.

    1. We are given that $50-p<0.25p$. Multiplying both sides by 4 gives $200-4p<p$, or $200<5p$, and hence $p>40$. However, we cannot determine whether $p>42$. For example, if $p=41$, then $p>40$ and $p<42$. On the other hand, if $p=43$, then $p>40$ and $p>42$; NOT sufficient.
    2. We are given that $50-p>0.1p$. Multiplying both sides by 10 gives $500-10p>p$, or $500>11p$, and hence $p<\frac{500}{11}=45.\overline{45}$ (long division; for what follows, it is enough to know that $\frac{500}{11}>45$).

       However, we cannot determine whether $p>42$. For example, if $p=41$, then $p<\frac{500}{11}=$ and $p<42$.

       On the other hand, if $p=43$, then $p<\frac{500}{11}$ and $p>42$; NOT sufficient.

    Taking (1) and (2) together, it is not possible to determine whether $p>42$ because the same examples used in (1) above are also used in (2) above.

    The correct answer is E; both statements together are still not sufficient.

40. An estimate of an actual data value has an error of p percent if $p=\frac{100\,|e-a|}{a}$, where e is the estimated value and a is the actual value. Emma's estimate for her total income last year had an error of less than 20 percent. Emma's estimate of her income from tutoring last year also had an error of less than 20 percent. Was Emma's actual income from tutoring last year at most 45 percent of her actual total income last year?

    1. Emma's estimated income last year from tutoring was 30 percent of her estimated total income last year.
    2. Emma's estimated total income last year was \$40,000.

    Arithmetic Estimation

    Given that Emma's estimates for both her total income and her income from tutoring last year, E_I and E_T, respectively, were within 20 percent of her actual total income and her actual income from tutoring, A_I and A_T, respectively, it follows that $0.8E_I<A_I<1.2E_I$ and $0.8E_T<A_T<1.2E_T$.

    Determine whether Emma's actual income from tutoring was at most 45 percent of her actual total income or if $\frac{A_T}{A_I}\le 0.45$.

    1. Given that $E_T=0.3E_I$, it follows from $0.8E_T<A_T<1.2E_T$ that $0.24E_I<A_T<0.36E_I$. Then, since $0.8E_I<A_I<1.2E_I$, it follows that $\frac{1}{1.2E_I}<\frac{1}{A_I}<\frac{1}{0.8E_I}$. Multiplying the inequalities gives

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$$\frac{0.24E_I}{1.2E_I} < \frac{A_T}{A_I} < \frac{0.36E_I}{0.8E_I} \text{ or } 0.2 < \frac{A_T}{A_I} < 0.45; \text{ SUFFICIENT.}$$

2. Given that Emma's estimated total income last year was $40,000, it is impossible to determine whether her actual income from tutoring was at most 45 percent of her actual total income because no information is given about her actual income from tutoring other than it was within 20 percent of her estimated income from tutoring. And there is no information from which her estimated income from tutoring can be determined; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

41. Was Store K's profit last month at least 10 percent greater than its profit the previous month?

    1. Store K's expenses last month were 5 percent greater than its expenses the previous month.
    2. Store K's revenues last month were 10 percent greater than its revenues the previous month.

Algebra Applied Problems

Let P_{last}, E_{last}, and R_{last} be, respectively, the profit, expenses, and revenues for last month. Also, let P_{previous}, E_{previous}, and R_{previous} be, respectively, the profit, expenses, and revenues for the previous month. Then we have $P_{\text{last}} = R_{\text{last}} - E_{\text{last}}$ and $P_{\text{previous}} = R_{\text{previous}} - E_{\text{previous}}$. Determine whether $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true, or equivalently, determine whether $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is true.

1. Given that $E_{\text{last}} = 1.05E_{\text{previous}}$, it follows that the inequality $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is equivalent to $R_{\text{last}} - 1.1R_{\text{previous}} \geq -0.05E_{\text{previous}}$. It is clear that, for suitable values of R_{last}, R_{previous}, and E_{previous}, this last inequality could be true and this last inequality could be false; NOT sufficient.
2. Given that $R_{\text{last}} = 1.1R_{\text{previous}}$, it follows that the inequality $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is equivalent to $E_{\text{last}} \leq 1.1E_{\text{previous}}$. It is clear that, for suitable values of E_{last} and E_{previous}, this last inequality could be true and this last inequality could be false; NOT sufficient.

Taking (1) and (2) together, the following shows that $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true.

$$\begin{aligned} P_{\text{last}} &= R_{\text{last}} - E_{\text{last}} \\ &= 1.1R_{\text{previous}} - 1.05E_{\text{previous}} \\ &\geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}} \end{aligned}$$

Therefore, $P_{\text{last}} \geq 1.1(R_{\text{previous}} - E_{\text{previous}})$, and hence $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true. (The reason for using $\geq$ above instead of $>$ is to allow for the possibility that $E_{\text{previous}} = 0$.)

The correct answer is C; both statements together are sufficient.

42. Yesterday, Diana spent a total of 240 minutes attending a training class, responding to emails, and talking on the phone. If she did no two of these three activities at the same time, how much time did she spend talking on the phone?

    1. Yesterday, the amount of time that Diana spent attending the training class was 90 percent of the amount of time that she spent responding to emails.
    2. Yesterday, the amount of time that Diana spent attending the training class was 60 percent of the total amount of time that she spent responding to emails and talking on the phone.

Arithmetic Percents

Let T, E, and P be the number of minutes, respectively, that Diana spent attending the training class, responding to emails, and talking on the phone. We are given that $T + E + P = 240$ and are to determine the value of P.

1. Given that $T = 0.9E$, it is not possible to determine the value of P. Note that more than one appropriate value is possible for E (for example, $E = 50$ and $E = 100$), and each of these values leads to a possible value of T (for example, $T = 0.9 \times 50 = 45$ and $T = 0.9 \times 100 = 90$), after which different possible values of P can be found (for example, $P = 240 - 50 - 45 = 145$ and $P = 240 - 100 - 90 = 50$);

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values of P can be found (for example, $P = 240 - 50 - 45 = 145$ and $P = 240 - 100 - 90 = 50$); NOT sufficient.

2. Given that $T = 0.6(E + P)$, it is not possible to determine the value of P. Note that using $T + E + P = 240$, we have $0.6(E + P) + (E + P) = 240$, or $1.6(E + P) = 240$, and hence $E + P = 150$. From this it follows that $T = 240 - (E + P) = 240 - 150 = 90$, but it is still not possible to determine the value of P since only $E + P = 150$ is required for the values of E and P. For example, it is possible that $T = 90$, $E = 50$, and $P = 100$, and it is possible that $T = 90$, $E = 100$, and $P = 50$; NOT sufficient.

Taking (1) and (2) together, we have $T = 0.9E$, $E + P = 150$, and $T = 90$. Therefore, $0.9E = 90$, or $E = 100$, and hence from $E + P = 150$, it follows that $100 + P = 150$, or $P = 50$.

The correct answer is C; both statements together are sufficient.

43.

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